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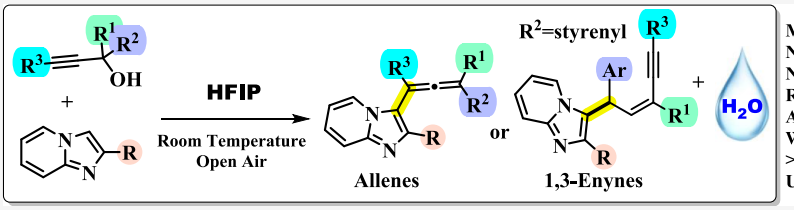
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HFIP-Promoted Friedel–Crafts Allenylation of Imidazopyridines with Propargyl Alcohols at Room Temperature

Srinivasarao Yaragorla,* Aayesha Shaik, and Dandugula Sneha Latha

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ABSTRACT: Herein, we report an efficient synthetic protocol for the Friedel–Crafts reaction of imidazo[1,2-*a*]pyridines with propargyl alcohols in HFIP. Notably, this FC allenylation works without any additional solvent or catalyst and requires neither inert conditions nor heating. In this method, HFIP, besides offering hydrogen bonding with propargyl alcohol, also stabilizes the resultant carbocation, thus forming the product. This method also enables the straightforward synthesis of 1,3-enynes. The current method offers a large substrate diversity with good to excellent yields. We also demonstrated gram-scale synthesis and synthetic modifications.

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INTRODUCTION

Among the diverse range of nitrogen-containing privileged heterocycles, imidazo[1,2-*a*]pyridines occupied a special place in synthetic organic and medicinal chemistry due to their rich presence as natural products and pharmaceuticals.[1] The derivatives of imidazo[1,2-*a*]pyridines exhibit a broad range of biological activities including, antiulcer,[2a] anti-inflammatory, anticonvulsant,[2b] antifungal,[2c] antiviral,[2d] anticancer, antiprotozoal,[2e] antibacterial,[3e] anthelmintic,[3b] analgesic,[3c] antitubercular,[3d] antiepileptic,[3e] and antitumor activities.[3f] Notably, there are marketed drugs such as saripidem,[4] alpidem,[5a] olprinone,[5b] zolimidine,[6] necopidem and zolpidem,[7] and rifaximin[8] that contain the imidazo[1,2-*a*]pyridine as the core moiety (Figure 1). Besides, these molecules are important in the preparation of polymer films due to their significant fluorescent properties.[9] Subsequently, the synthesis and derivatization of imidazo[1,2-*a*]pyridines, in particular, C3-functionalization, received an overwhelming response from many research groups, toward the development of C–H alkylation, arylation, aroyl methylation, hydroxy-alkylation, thio-carbamation, aminomethylation, sulfonylation, thio-cyanation, selenization, halogenation, amination, trifluoromethylation, sulfenylation, carbonylation, selenocyanation and many more.[10,11] Apparently, electrophilic aromatic substitution, and, in particular, Friedel–Crafts alkylation or acylation, is probably one of the most recurrent tools used to derivatize imidazo[1,2-*a*]pyridines using conventional metal-catalyzed strategies. For example, Hajra and coworkers reported the FC reaction of imidazopyridines and propargyl alcohols by refluxing them in toluene at 110 °C for 16 h with $Zn(OTf)_2$ as the catalyst.[12]

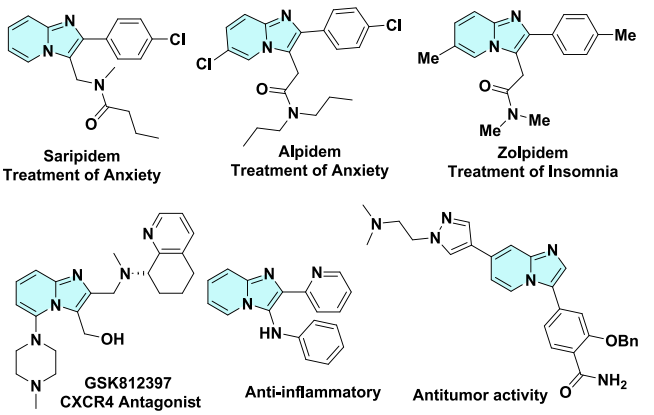


Figure 1. Representative examples of biologically active and marketed drugs with C3-functionalized imidazo[1,2-*a*]pyridine core structure.

On the other hand, 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP), the magic solvent, proved to be an excellent promoter system in organic synthesis, particularly for the carbon–carbon bond formation reactions.[13] Although the role of HFIP is initially realized in stabilizing the cationic intermediates,

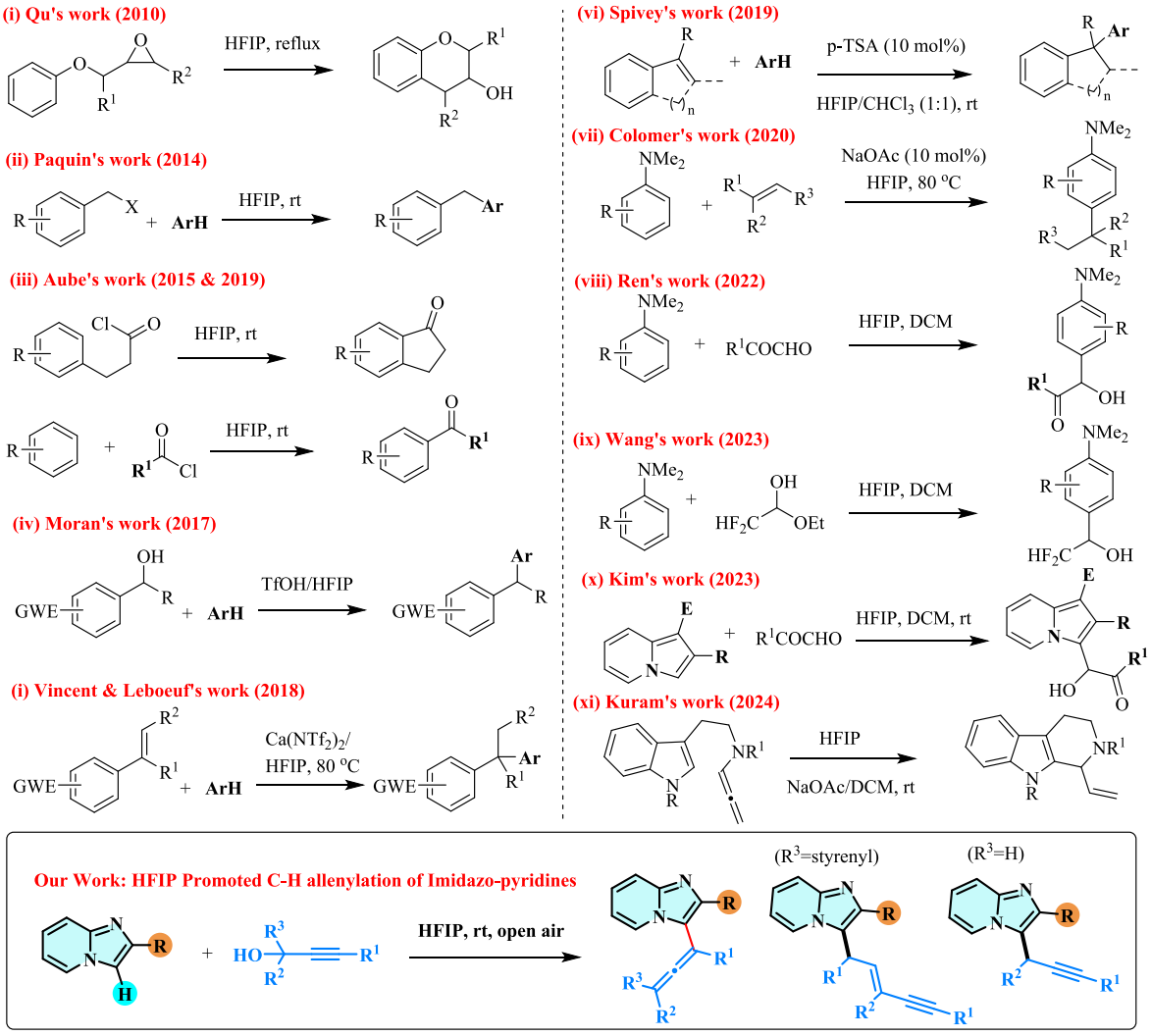
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Scheme 1. State-of-the-Art of Friedel-Crafts Reactions in HFIP



Berkessel's group demonstrated the role of the hydrogen bonding ability of HFIP in promoting the epoxidation of alkenes.[14] HFIP facilitates electrophilic aromatic substitution reactions due to its peculiar properties, such as higher polarity, increased Brønsted acidity, high ionizing power, low nucleophilicity, and strong hydrogen-bond donation properties. Therefore, a number of investigators have studied FC alkylation chemistry in HFIP. For example, Cativiela et al. reported that FC alkylation product 3-(2-furyl)-propanal was formed as a byproduct during the Diels−Alder reactions of furan and acrolein in HFIP.[15] Subsequently, a few more elegant approaches toward FC alkylations have been reported from allylic alcohols,[16] epoxides,[17] and benzyl halides,[18] intramolecular benzoylation,[19] acylation,[20] π-activated alcohols,[21] styrenes,[22] *N*,*N*-dialkyl amines,[23] indolizines[24] and indole tethered allenes[25] in HFIP (Scheme 1).

Recently, we reported FC-arylation of various α-imino ketones and subjected them to cyclization to obtain 2-aryl pyrroles and 2-aryl indoles.[26] We also demonstrated that HFIP can promote the ring-opening cyclization of DA-cyclopropanes to obtain 2,3-dihydro pyrroles and benzimidazoles.[27a,b] Later, a mechanochemical approach for the formal [3 + 3] cyclization of various enaminones with propargyl alcohols was also developed with HFIP.[27c] Given the importance of the FC reactions in HFIP, we sought to develop the allenylation of imidazo[1,2-*a*]pyridines with propargyl alcohols in the presence of HFIP and report our initial results here.

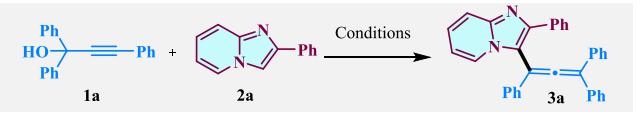
RESULTS AND DISCUSSION

To investigate the FC reaction conditions, we chose propargyl alcohol (**1a**) and imidazo[1,2-*a*]pyridine (**2a**) as the suitable reacting partners (Table 1). Initially, a mixture of **1a** (0.52 mmol), **2a** (0.63 mmol) and 5/5 mol % $Ca(OTf)_2/Bu_4NPF_6$ was stirred in HFIP at room temperature for 24 h and to our delight, we observed the desired product was formed in less than 76% (entry 1, Table 1). Increasing the catalyst loading was undesirable as there was no improvement in the yield (entry 2). To check the role of HFIP, we performed the reaction without HFIP and realized that the product was formed in a trace (entry 3). This indicated that the HFIP solely promoted the reaction. Hence, we performed the reactions with HFIP, by varying its volume from 0.1 to 0.3 mL and finally, 0.5 mL was found to be excellent for yielding **3a** in 93% within 15 min at room temperature (entries 4−6).

Having the optimum reaction conditions in hand, we then switched to studying the generality of the FC reaction. As showcased in Scheme 2, initially, we studied the scope of various propargyl alcohols in the FC reaction of imidazo[1,2-

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Table 1. Optimization of Reaction Conditions[a]



entry	reaction conditions	yield (%)[b]
1	$Ca(OTf)_2/Bu_4NPF_6$ (5/5 mol %), HFIP (0.1 mL), rt, 24 h	76
2	$Ca(OTf)_2/Bu_4NPF_6$ (10/10 mol %), HFIP (0.1 mL), rt, 18 h	78
3	$Ca(OTf)_2/Bu_4NPF_6$ (5/5 mol %), neat, rt, 24 h	trace
4	HFIP (0.1 mL), rt, 24 h	45
5	HFIP (0.3 mL), rt, 2.5 h	80
6[c]	HFIP (0.5 mL), rt, 15 min	93

[a]Unless mentioned, all the reactions were carried out with **1a** (150 mg, 0.52 mmol) and **2a** (123 mg, 0.63 mmol). [b]Isolated yields. [c]Optimum condition.

a]pyridine (**2a**). Under the standard conditions, propargyl alcohols, **1a** to **1p**, furnished the corresponding products **3a** to **3p** in excellent yields. These examples illustrate that electronic and steric factors on propargyl alcohols did not remarkably influence product yields. This method allowed the synthesis of not only tetra aryl allenes but also alkyl-substituted allenes (**3j**−**3l**, **3ad**). Next, we turned our focus to studying the scope of substitutions on the pyridine ring of **2** and synthesized the corresponding allenes **3q**−**3s** in good yields. The substitutions on the benzene ring of 2-aryl-imidazo-pyridines were also found to be less influenced by the electronic effects and provided the desired compounds (**3t**−**3w**) with excellent yields. Examples **3y** to **3ac** indicate that the protocol tolerates the diversity on both the substrates **1** and **2**. Further, it is observed that simple imidazo[1,2-*a*]pyridine (**2y**, C2-unsubstituted) did not react with **1a** under standard conditions for 24 h. When heated to 60 °C (oil bath temperature) for 24 h, a trace amount of product was observed, which was confirmed by HRMS data (see Supporting Information). The FC reaction with other arenes such as furan and pyrrole was not successful at rt and heating.

As indicated in Scheme 3, oxindole-derived, *tert*-propargyl alcohol (**4**) did not react with **2j** under standard reaction conditions even after 24 h. Heating the reaction mixture at 60 °C was also not desirable as we did not observe any reaction. Next, *sec*-propargyl alcohol (**5**) reacted with **2j** and through a direct FC benzylation to furnish the compound **6** in 85% yield.

Encouraged by these observations, we then switched to studying the FC reaction of **2** with vinyl propargyl alcohol **7** (Scheme 4). Interestingly, substrate **7** has competitive functional groups, such as allyl alcohol and propargyl alcohol, so we can study the chemoselectivity of FC reaction. To our delight, propargyl alcohol **7a** reacted smoothly with various imidazopyridines in HFIP at room temperature and furnished the corresponding FC products **8a** to **8e** in good yields. Notably, this turned out to be an excellent and simplest method for the synthesis of 1,3-enynes. Notably, the FC arylation occurred on allyl alcohol in preference to the propargyl alcohol via S_N2' mechanism. Similarly, **7b** and **7c** also reacted with imidazopyridines and furnished the desired compounds **8f** and **8g** (Scheme 4). Further, the FC arylation of these vinyl propargyl alcohols was extended to a different arene, *N*,*N*-dimethyl aniline by treating it with **7d** in HFIP at rt for 30 min, to obtain the desired compound **9** in 87% yield (Scheme 5).

Having studied the generality of the FC reaction with various propargyl alcohols, we demonstrated the gram-scale synthesis of **3a** from **1a** and **2a** and synthesized 1.4 g of **3a** (Scheme 6). Besides, we demonstrated the synthetic potential of the products by treating **3a** with NIS in acetonitrile at rt for 6 h and confirmed that the allene converted into vinyl iodide **10** in 68% yields. Similarly, **3s** also furnished the vinyl iodide **11**. We have provided the possible route for forming vinyl iodide from allene. We also reasoned the formation of vinyl iodides. Notably, the two double bonds of the allene are distinguished by the higher electron density on the imidazopyridine-substituted alkene; hence, it underwent cyclic iodonium formation, followed by the nucleophilic opening with water furnished the vinyl iodides **10**/**11** as indicated in Scheme 6.

After successfully demonstrating the protocol, we proposed the reaction mechanism for this protocol in Scheme 7. We believe that the activation of propargyl alcohol **1** with HFIP proceeds as indicated in structure **A** to furnish the propargyl cation **B**. Isomerization of **B** to allenyl cation (**C**) triggers the electrophilic aromatic substitution (**D**), (Friedel−Crafts) reaction to furnish the desired compound **3**. Alternatively, path-b can not be ignored, as the arene **2** can directly react with activated propargyl alcohol to form **3** via the transition model (**i**). In the case of the FC reaction of vinyl-propargyl alcohol (**7**) with arene **2**, we proposed that the reaction proceeds as depicted in the model (**ii**). Likewise, the FC reaction of *sec*-propargyl alcohols **5** with **2** may proceed via the model (**iii**, Scheme 7).

CONCLUSION

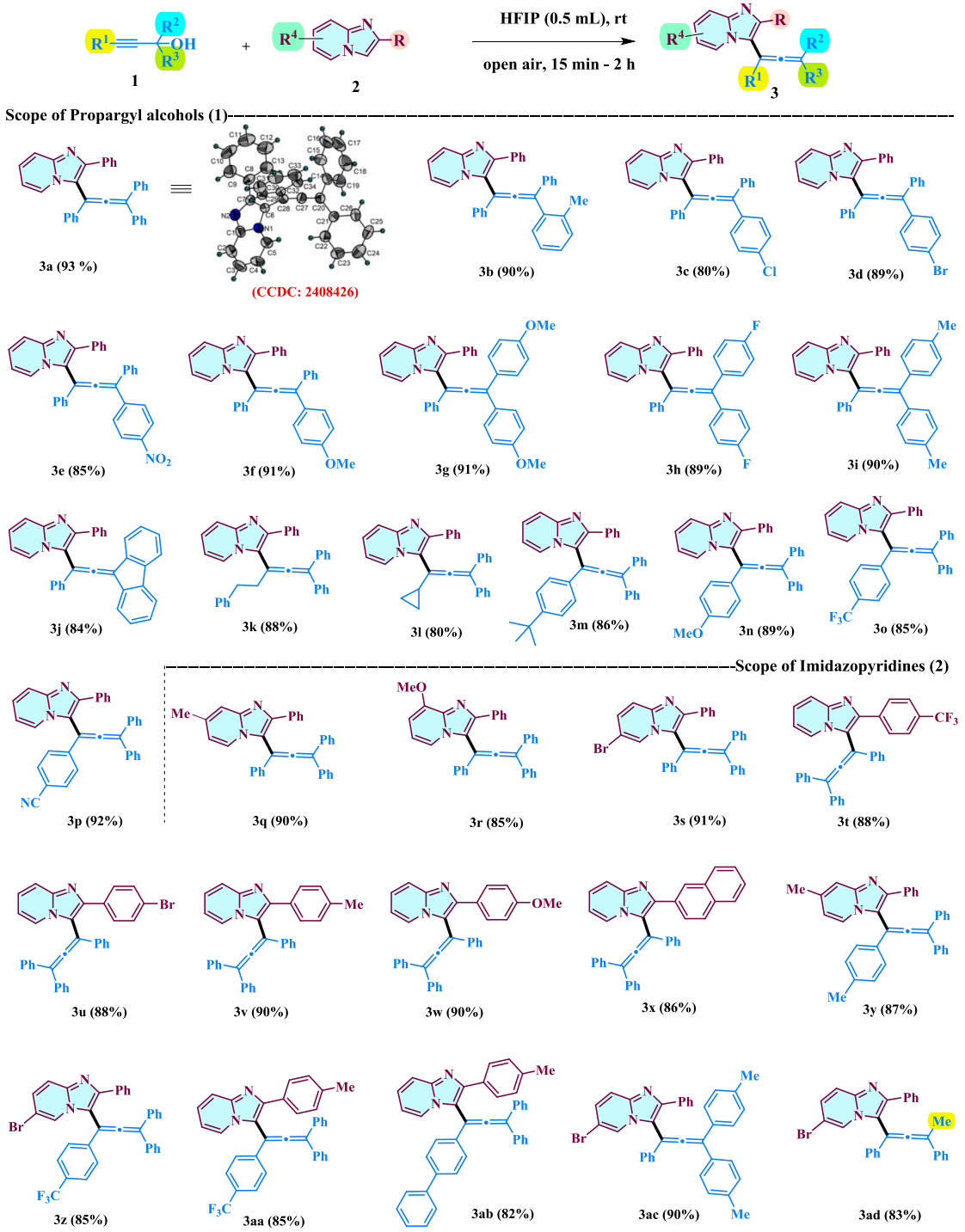
In summary, we have developed a highly facile synthetic protocol for the Friedel−Crafts reaction of imidazopyridine with various propargyl alcohols in HFIP. We demonstrated that allenes could be synthesized from imidazopyridine and *tert*-propargyl alcohols and alkylation with sec-propargyl alcohols. Besides, vinyl-propargyl alcohols provided the efficient synthesis of 1,3-enynes through FC-alkylation. All these reactions were carried out without any additional catalyst or solvent. The hydrogen bonding ability and the stabilization of the carbocation by the HFIP might probably be the key driving force that facilitates these transformations. The reaction offers excellent substrate scope and yields. Besides, we demonstrated the gram scale synthesis and postsynthetic modifications of the products. We believe that this unique application for the synthesis of allenes through FC reaction promoted by HFIP opens up many such novel developments within its domain of the field.

EXPERIMENTAL SECTION

General. Unless otherwise noted, all reagents were used as received from commercial suppliers. All reactions were performed in a 5 mL open RB flask at room temperature. Reactions were monitored using thin-layer chromatography (TLC) with aluminum sheets of silica gel 60 F_{254} from Merck. TLC plates were visualized with UV light (254 nm), iodine treatment or using *p*-anisaldehyde stain. Column chromatography was carried out using silica gel 100−200 mesh as the stationary phase. NMR spectra were recorded at 500 MHz and (^{1}H) and at 125 and 100 MHz (^{13}C), respectively, on the

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Scheme 2. Substrate Scope[a]



[a](Reaction conditions: A mixture of 1 (150 mg, 1 equiv), 2 (1.2 equiv) and HFIP (0.5 mL) was stirred at room temperature in an open flask).

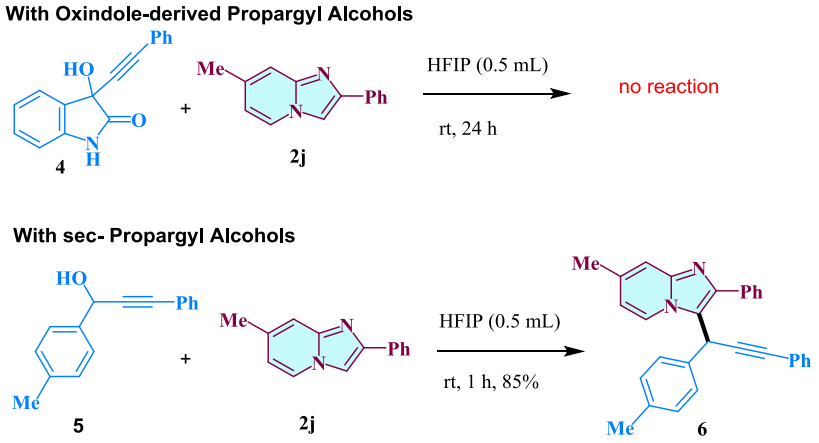
Avance Bruker spectrometer. Chemical shifts (δ) are reported in ppm, using the residual solvent peak in $CDCl_3$ (^1H: δ = 7.26 and ^{13}C: δ = 77.16 ppm) as an internal standard, and coupling constants (J) are given in Hz. HRMS were recorded using ESI-TOF techniques. Melting points were measured with the LABINDIA mepa melting apparatus. Single crystal X-ray diffraction data were collected in Bruker D8-Quest, diffractometers.

General Experimental Procedure for the Synthesis of ***3****.* A mixture of propargyl alcohol **1** (150 mg, 1 equiv), imidazo[1,2-*a*]pyridine **2** (1.2 equiv) in HFIP (0.5 mL) was stirred at room temperature for 15 min to 2 h. After completion of the reaction (monitored by TLC), the resulting crude product was purified by column chromatography (15% EtOAc in petroleum ether) to obtain the corresponding allene **3** in good yield.

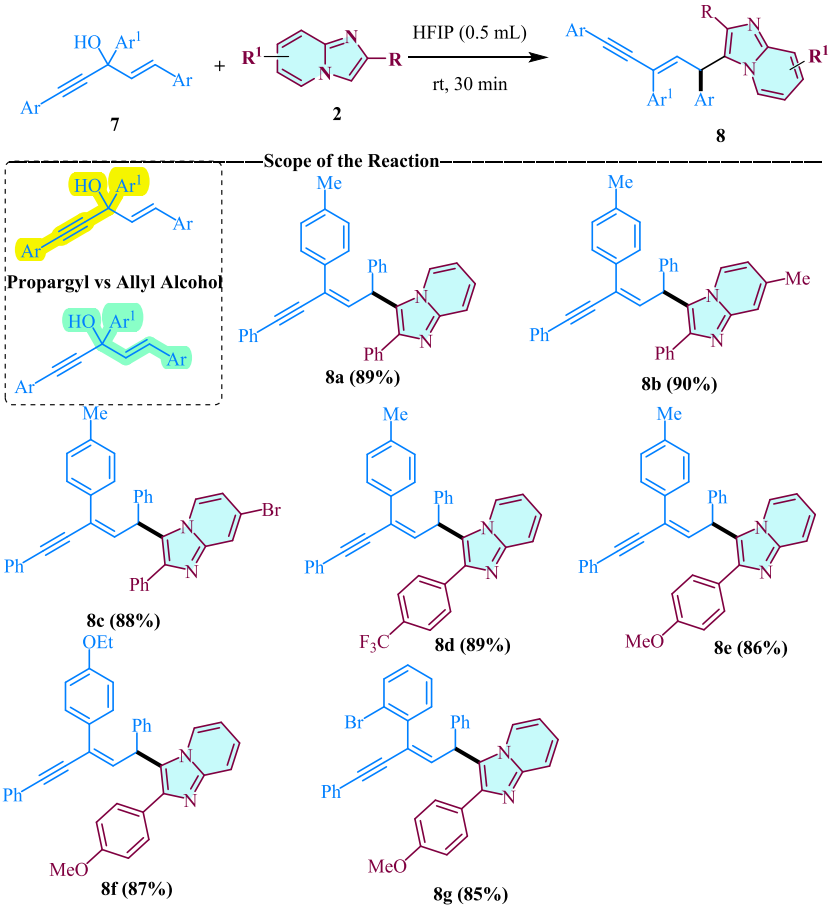
General Experimental Procedure for the Synthesis of ***6*** *and* ***8****.* A mixture of *sec*-propargyl alcohol **5** (1 equiv), imidazo[1,2-*a*]pyridine **2** (1.2 equiv) in HFIP (0.5 mL) was stirred at room temperature for 1 h. After completion of the reaction (monitored by TLC), the resulting crude product was

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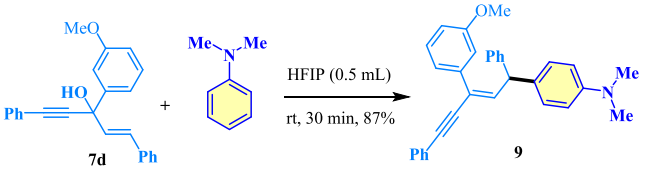
Scheme 3. FC-Reaction of 4 and 5 in HFIP



Scheme 4. Chemoselective FC-Reaction for the Synthesis of 1,3-Enynes



Scheme 5. FC Arylation of 7d with *N,N*-Dimethyl Aniline



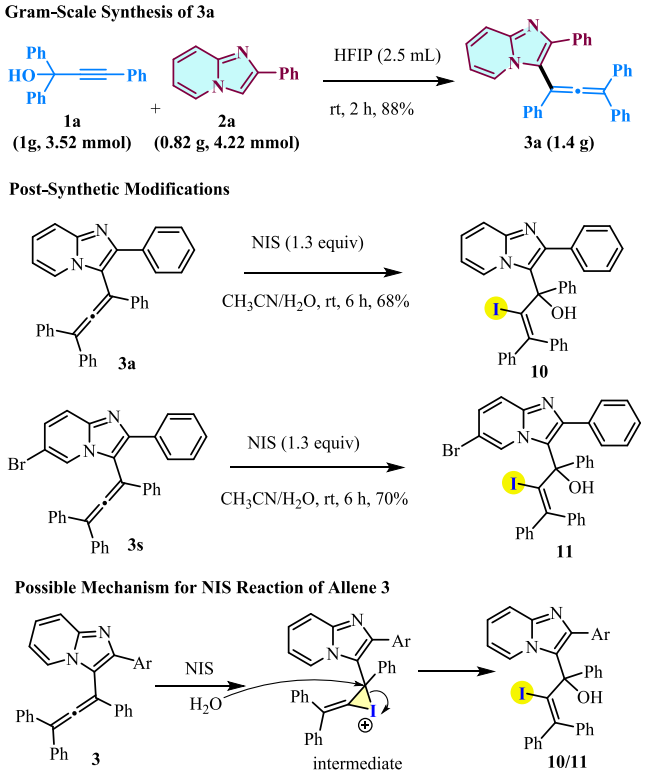
purified by column chromatography (30% EtOAc in petroleum ether) to obtain the product **6** in 85% yield. Under similar conditions, propargyl alcohol **7** and **2** reacted to furnish the desired compounds **8** in good yields.

*General Experimental Procedure for the Synthesis of **10** and **11**.* A mixture of allene **3a** (1 equiv), *N*-iodosuccinimide (1.3 equiv) was stirred in MeCN (1 mL) at room temperature for 6 h. After completion of the reaction (monitored by TLC), the resulting crude product was purified by column chromatography (50% EtOAc in petroleum ether) to obtain the vinyl iodide **10** in 68% yield. Using the same procedure, allene **3s** furnished the vinyl iodide **11** in 70% yield.

Spectral Data. *2-Phenyl-3-(1,3,3-triphenylpropa-1,2-dien-1-yl)imidazo[1,2-a] pyridine **3a**.*[12] Following the general experimental procedure **I**, compound **3a** was obtained (using 15% EtOAc/petroleum ether) as white solid (226 mg, 93%);

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Scheme 6. Demonstration of Gram-Scale Synthesis and Post-Synthetic Modifications

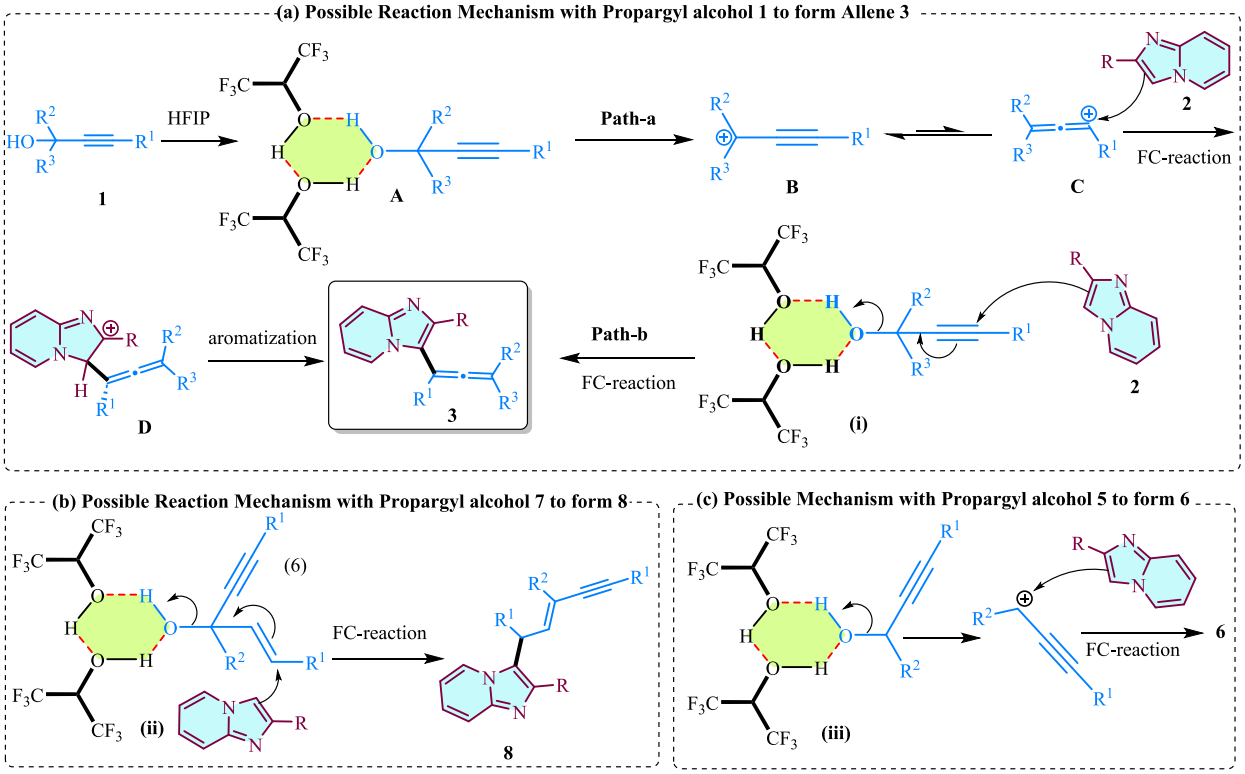


mp: 124−125 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.89−7.87 (m, 2H), 7.79−7.78 (m, 1H), 7.71 (d, J = 9.00 Hz, 1H), 7.36−7.34 (m, 2H), 7.32−7.28 (m, 3H), 7.28−7.26 (m, 5H), 7.25−7.21 (m, 7H), 7.20−7.18 (m, 2H), 6.67−6.64 (m, 1H) ppm; ^{13}C {^{1}H} NMR (125 MHz, $CDCl_3$): δ 209.8, 145.4, 143.9, 135.3, 133.9, 133.8, 129.3, 128.8, 128.7, 128.4, 128.2, 128.1, 128.0, 127.7, 126.3, 124.9, 124.3, 117.6, 114.3, 113.9, 112.5, 100.8 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{25}N_2]^+$: 461.2018; found: 461.2019.

3-(1,3-Diphenyl-3-(o-tolyl) propa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a] pyridine ***3b***. Following the general experimental procedure **I**; compound **3b** was obtained (using 10% EtOAc/petroleum ether) as brown gummy product (213 mg, 90%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.81 (d, J = 7.15 Hz, 2H), 7.75 (d, J = 6.80 Hz, 1H), 7.69 (d, J = 9.00 Hz, 1H), 7.34−7.28 (m, 4H), 7.26−7.20 (m, 6H), 7.18−7.12 (m, 4H), 7.10−7.07 (m, 4H), 6.67−6.64 (m, 1H), 1.93 (s, 3H) ppm; ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 208.3, 145.4, 144.0, 137.0, 135.2, 134.0, 133.9, 133.6, 130.5, 130.4, 129.3, 128.8, 128.2 (2), 128.1, 127.8, 127.7, 127.2, 126.4, 126.1, 125.0, 124.3, 117.5, 114.3, 112.5, 112.0, 100.3, 20.2 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{27}N_2]^+$: 475.2174; found: 475.2172.

3-(3-(4-Chlorophenyl)-1,3-diphenylpropa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a]pyridine ***3c***. Following the general experimental procedure **I**, compound **3c** was obtained (using 10% EtOAc/petroleum ether) as white solid (186 mg, 80%); mp: 243−244 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.84−7.82 (m, 2H), 7.76 (d, J = 6.85 Hz, 1H), 7.70 (d, J = 9.05 Hz, 1H), 7.33−7.32 (m, 4H), 7.31−7.27 (m, 4H), 7.25−7.23 (m, 3H), 7.21−7.19 (m, 5H), 7.12−7.10 (m, 2H), 6.69−6.66 (m, 1H) ppm; ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 209.7, 145.4, 143.9, 137.5, 134.8, 133.9, 133.7, 133.6, 133.5, 130.3, 130.0, 129.4, 128.9, 128.8, 128.6, 128.5, 128.4, 128.3, 128.0, 127.9, 127.5, 126.3, 125.1, 124.1, 117.6, 114.1, 112.9, 112.7, 101.0

Scheme 7. Possible Reaction Mechanism



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ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{24}ClN_2]^+$: 495.1628; found: 495.1632.

3-(3-(4-Bromophenyl)-1,3-diphenylpropa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a]pyridine ***3d***. Following the general experimental procedure **I**, compound **3d** was obtained (using 10% EtOAc/petroleum ether) as white solid (198 mg, 89%); mp: 243−244 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.70−7.68 (m, 3H), 7.62−7.60 (m, 1H), 7.27−7.24 (m, 5H), 7.19−7.17 (m, 5H), 7.16−7.09 (m, 6H), 6.95 (d, J = 8.45 Hz, 2H), 6.63 (t, J = 6.88 Hz, 1H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.7, 145.3, 143.7, 134.7, 134.1, 133.3, 133.1, 131.8, 130.3, 129.5, 128.8, 128.6 (2), 128.5, 128.3, 128.1, 126.2, 125.6, 124.2, 122.1, 117.2, 114.2, 113.0, 100.8 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{24}BrN_2]^+$: 539.1123; found; 539.1120 and $[M+H+2]^+$ calculated for $[C_{34}H_{25}BrN_2]^{2+}$: 540.1190; found; 540.1187; IR (film): ν_{max} 1594, 1487, 1400, 1234, 1071, 1026, 875, 729, 691 cm^{-1}.

3-(3-(4-Nitrophenyl)-1,3-diphenylpropa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a]pyridine ***3e***. Following the general experimental procedure **I**, compound **3e** was obtained (using 10% EtOAc/petroleum ether) as brown solid (196 mg, 85%); mp: 156−157 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.05 (d, J = 7.95 Hz, 2H), 7.80−7.76 (m, 3H), 7.71 (d, J = 8.90 Hz, 1H), 7.37−7.36 (m, 4H), 7.31−7.27 (m, 6H), 7.25−7.22 (m, 4H), 7.18−7.16 (m, 2H), 6.72 (t, J = 6.75 Hz, 1H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): $\underline{\delta}$ 210.4, 147.3, 145.7, 144.4, 142.3, 134.1, 133.5, 132.9, 129.7, 129.3, 129.1, 128.9, 128.6 (2), 128.1 (2), 126.4, 125.3, 124.1, 123.9, 117.7, 113.4, 113.0, 112.4, 101.9 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{24}N_3O_2]^+$: 506.1869; found: 506.1872; IR (film): ν_{max} 2922, 2852, 1597, 1517, 1342, 1123, 879, 743, 695 cm^{-1}.

3-(3-(4-Methoxyphenyl)-1,3-diphenylpropa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a] pyridine ***3f***. Following the general experimental procedure **I**, compound **3f** was obtained (using 10% EtOAc/petroleum ether) as white gummy product (213 mg, 91%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.85 (dd, J = 1.50, 2.00 Hz, 2H), 7.79 (d, J = 6.85 Hz, 1H), 7.69 (d, J = 9.05 Hz, 1H), 7.34−7.28 (m, 5H), 7.26−7.23 (m, 5H), 7.22−7.18 (m, 4H), 7.15−7.14 (m, 2H), 6.80−6.78 (m, 2H), 6.66 (t, J = 6.75 Hz, 1H), 3.79 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.7, 159.6, 145.3, 143.5, 135.5, 134.0, 133.5, 130.0, 129.3, 128.8, 128.7, 128.5, 128.1 (2), 127.8, 127.2, 126.2, 125.2, 124.3, 117.3, 114.7, 114.2, 113.6, 112.7, 100.2, 55.4 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{27}N_2O]^+$: 491.2123; found: 491.2120; IR (film): ν_{max} 3273, 3043, 2063, 1610, 1515, 1241, 1031, 839, 765, 697 cm^{-1}.

3-(3,3-Bis(4-methoxyphenyl)-1-phenylpropa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a] pyridine ***3g***. Following the general experimental procedure **I**, compound **3g** was obtained (using 10% EtOAc/petroleum ether) as white solid (206 mg, 91%); mp: 141−142 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 6.97−6.95 (m, 2H), 6.71 (d, J = 9.05 Hz, 1H), 6.64 (d, J = 6.80 Hz, 1H), 6.49−6.48 (m, 2H), 6.47−6.46 (m, 5H), 6.42 (s, 2H), 6.35−6.31 (m, 1H), 6.24−6.19 (m, 4H), 5.78−5.73 (m, 4H), 2.87 (s, 3H), 2.86 (s, 3H) ppm; $^{13}C\{^1H\}$NMR (125 MHz, $CDCl_3$): δ 207.3, 145.2, 141.4, 140.7, 136.2, 133.4, 131.7, 129.9, 129.0, 128.7, 128.6, 128.5, 128.4, 128.2, 128.0, 127.5, 126.3, 126.2, 124.9, 124.4, 122.2, 117.6, 116.6, 112.5, 112.4, 97.9, 34.8, 34.1 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{36}H_{29}N_2O_2]^+$: 521.2229; found: 521.2226.

3-(3,3-Bis(4-fluorophenyl)-1-phenylpropa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a]pyridine ***3h***. Following the general experimental procedure **I**, compound **3h** was obtained (using 10% EtOAc/petroleum ether) as white solid (206 mg, 89%); mp: 165−166 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.84−7.83 (m, 2H), 7.74 (dt, J = 6.85, 1.05 Hz, 1H), 7.70 (dt, J = 9.05, 0.95 Hz, 1H), 7.33 (d, J = 4.35 Hz, 4H), 7.31−7.29 (m, 1H), 7.24−7.19 (m, 4H), 7.16−7.13 (m, 4H), 6.95−6.92 (m, 4H), 6.68 (td, J = 6.80, 0.95 Hz, 1H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.3, 163.8, 161.4, 145.5, 144.1, 133.7, 133.5, 131.1, 130.3, 130.2, 129.5, 128.5, 128.4, 128.0, 127.8, 126.2, 125.0, 124.1, 117.7, 115.9, 115.6, 114.0, 112.7, 111.9, 101.0 ppm; ^{19}F (470 MHz, $CDCl_3$): −113.4 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{23}F_2N_2]^+$: 497.1829; found: 497.1825; IR (film): ν_{max} 1767, 1599, 1504, 1448, 1230, 1201, 1158, 835, 694 cm^{-1}.

2-Phenyl-3-(1-phenyl-3,3-di-p-tolylpropa-1,2-dien-1-yl)-imidazo[1,2-a]pyridine ***3i***. Following the general experimental procedure **I**, compound **3i** was obtained (using 10% EtOAc/petroleum ether) as white solid (211 mg, 90%); mp: 182−183 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.91−7.89 (m, 2H), 7.80−7.79 (m, 1H), 7.72 (d, J = 8.90 Hz, 1H), 7.34−7.27 (m, 4H), 7.23−7.20 (m, 5H), 7.14−7.12 (m, 4H), 7.07−7.06 (m, 4H), 6.66 (t, J = 6.75 Hz, 1H), 2.34 (s, 6H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.7, 145.2, 143.4, 138.0, 134.0, 133.6, 132.3, 129.4, 129.3, 128.7, 128.4, 128.0, 127.7, 126.2, 125.1, 124.3, 117.5, 114.6, 113.8, 112.6, 100.2, 21.3 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{36}H_{29}N_2]^+$: 489.2331; found: 489.2333; IR (film): ν_{max} 2918, 2850, 1632, 1505, 1444, 1264, 1180, 1022, 877, 763, 733, 692, 628 cm^{-1}.

3-(2-(9H-Fluoren-9-ylidene)-1-phenylvinyl)-2-phenylimidazo[1,2-a]pyridine ***3j***. Following the general experimental procedure **I**, compound **3j** was obtained (using 10% EtOAc/petroleum ether) as yellow gummy product (204 mg, 84%); ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.10 (d, J = 6.40 Hz, 1H), 7.88−8.7 (m, 2H), 7.72−7.69 (m, 3H), 7.41−7.39 (m, 2H), 7.35−7.34 (m, 2H), 7.31 (m, 3H), 7.3 (m, 5H), 7.24 (m, 1H), 7.17 (t, J = 7.20 Hz, 2H), 6.86 (t, J = 6.80 Hz, 1H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 207.4, 145.6, 144.9, 139.3, 136.6, 133.7, 133.1, 129.5, 128.8, 128.7 (2), 128.2, 127.3, 126.8, 125.6, 124.2, 123.6, 120.4, 117.5, 113.9, 113.3, 109.0, 105.5 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{23}N_2]^+$: 458.1783; found: 458.1785.

2-Phenyl-3-(1,1,5-triphenylpenta-1,2-dien-3-yl)imidazo-[1,2-a]pyridine ***3k***. Following the general experimental procedure **I**, compound **3k** was obtained (using 10% EtOAc/petroleum ether) as the brown gummy product (206 mg, 88%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.93−7.91 (m, 2H), 7.61 (dt, J = 9.00, 1.05 Hz, 1H), 7.54 (dt, J = 6.95, 1.10 Hz, 1H), 7.38−7.36 (m, 1H), 7.35−7.34 (m, 2H), 7.33−7.31 (m, 6H), 7.24−7.22 (m, 4H), 7.13−7.12 (m, 3H), 7.11−7.09 (m, 1H), 6.93−6.91 (m, 2H), 6.46 (td, J = 6.80, 1.20 Hz, 1H), 2.80−2.76 (m, 4H) ppm. $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 207.1, 145.0, 142.4, 140.8, 136.1, 134.3, 128.9, 128.6 (2), 128.4, 128.0, 127.9, 126.1, 124.7, 124.4, 117.3, 116.0, 112.2, 112.1, 98.0, 34.6, 34.0 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{36}H_{29}N_2]^+$: 489.2331; found: 489.2332; IR(film): ν_{max} 1632, 1495, 1445, 1340, 1263, 1097, 906, 802, 768, 694 cm^{-1}.

3-(1-Cyclopropyl-3,3-diphenylpropa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a]pyridine ***3l***. Following the general experimental procedure **I**, compound **3l** was obtained (using 10% EtOAc/petroleum ether) as brown gummy product (202 mg, 80%); ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.13 (d, J = 6.28

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Hz, 1H), 7.86−7.79 (m, 1H), 7.75−7.71 (m, 2H), 7.52−7.48 (m, 1H), 7.34−7.32 (m, 2H), 7.26−7.23 (m, 1H), 7.09 (m, 2H), 7.07 (m, 1H), 6.80−6.78 (m, 3H), 6.61−6.59 (m, 2H), 6.48−6.46 (m, 1H), 6.40−6.37 (m, 1H), 5.90−5.86 (m, 1H), 5.41−5.36 (m, 1H), 3.58−3.55 (m, 2H), 3.13−3.08 (m, 2H),ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.8, 144.9, 142.5, 136.1, 133.8, 130.2, 128.8, 128.6, 128.4 (2), 127.9 (2), 127.8 (2), 127.7, 126.5, 125.1, 124.3, 117.2, 116.6, 115.7, 113.4, 112.6, 101.8, 13.6, 7.9, 1.1 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{31}H_{25}N_2]^+$: 425.2018; found: 425.2015.

3-(1-(4-(tert-butyl)phenyl)-3,3-diphenylpropa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a]pyridine ***3m***. Following the general experimental procedure **I**, compound **3m** was obtained (using 10% EtOAc/petroleum ether) as white solid (195 mg, 86%); mp: 206−207 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.92 (d, J = 7.50 Hz, 2H), 7.86 (d, J = 6.80 Hz, 1H), 7.76 (d, J = 9.00 Hz, 1H), 7.37−7.35 (m, 2H), 7.32−7.29 (m, 6H), 7.28−7.27 (m, 8H), 7.25−7.21 (m, 2H), 6.71 (t, J = 6.70 Hz, 1H), 1.32 (s, 9H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 209.6, 151.4, 145.3, 143.6, 135.4, 133.7, 130.7, 128.8, 128.6, 128.4, 128.1, 128.0, 127.7, 126.3, 126.0, 125.0, 124.4, 117.5, 114.5, 113.7, 112.5, 100.4, 34.7, 31.3 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{38}H_{33}N_2]^+$: 517.2644; found: 517.2645.

3-(1-(4-Methoxyphenyl)-3,3-diphenylpropa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a] pyridine ***3n***. Following the general experimental procedure **I**, compound **3n** was obtained (using 15% EtOAc/petroleum ether) as yellow gummy product (209 mg, 89%); ^1H NMR (500 MHz, $CDCl_3$): δ 7.90−7.89 (m, 2H), 7.79 (d, J = 6.80 Hz, 1H), 7.71 (d, J = 9.00 Hz, 1H), 7.26−7.24 (m, 8H), 7.22−7.17 (m, 8H), 6.83 (d, J = 8.75 Hz, 2H), 6.65 (t, J = 6.80 Hz, 1H), 3.76 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 209.2, 159.8, 145.4, 143.6, 135.6, 133.8, 128.8, 128.7, 128.4, 128.3, 128.1, 128.0, 127.7, 127.6, 125.9, 124.9, 124.3, 117.6, 114.8, 114.6, 113.8, 112.5, 100.3, 55.4 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{27}N_2O]^+$: 491.2123; found: 491.2120.

3-(3,3-Diphenyl-1-(4-(trifluoromethyl)phenyl)propa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a]pyridine ***3o***. Following the general experimental procedure **I**, compound **3o** was obtained (using 20% EtOAc/petroleum ether) as brown solid (192 mg, 85%); mp: 160 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.86 (d, J = 1.40 Hz, 1H), 7.84 (d, J = 1.90 Hz, 1H), 7.74−7.71 (m, 2H), 7.55 (d, J = 8.40 Hz, 2H), 7.44 (d, J = 8.45 Hz, 2H), 7.31−7.28 (m, 5H), 7.25−7.22 (m, 7H), 7.21−7.19 (m, 2H), 6.67 (t, J = 6.80 Hz, 1H), ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 210.7, 145.6, 144.2, 137.9, 134.7, 133.6, 128.8 (2), 128.5, 128.4, 127.9, 126.5, 126.3 (2), 125.1, 124.0, 117.8, 114.5, 113.4, 112.7, 100.0 ppm; ^{19}F (376 MHz, $CDCl_3$): −62.55; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{24}F_3N_2]^+$: 529.1892; found: 529.1888.

4-(3,3-Diphenyl-1-(2-phenylimidazo[1,2-a]pyridin-3-yl)-propa-1,2-dien-1-yl)benzonitrile ***3p***. Following the general experimental procedure **I**, compound **3p** was obtained (using 20% EtOAc/petroleum ether) as the brown gummy product (216 mg, 92%); ^1H NMR (500 MHz, $CDCl_3$): δ 7.79−7.77 (m, 2H), 7.75−7.71 (m, 2H), 7.58−7.56 (m, 2H), 7.43−7.41 (m, 2H), 7.35−7.29 (m, 5H), 7.26−7.19 (m, 9H), 6.69 (t, J = 6.80 Hz, 1H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 211.2, 145.6, 144.0, 139.0, 134.3, 133.1, 128.9, 128.8, 128.6 (2), 128.2, 128.0, 126.8, 125.6, 123.9, 118.7, 117.6, 115.0, 113.1, 111.6, 99.9 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{24}N_3]^+$: 486.1970; found: 486.1976.

7-Methyl-2-phenyl-3-(1,3,3-triphenylpropa-1,2-dien-1-yl)-imidazo[1,2-a]pyridine ***3q***.[12] Following the general experimental procedure **I**, compound **3q** was obtained (using 15% EtOAc/petroleum ether) as white solid (226 mg, 90%); mp: 181−182 °C ^1H NMR (500 MHz, $CDCl_3$): δ 7.81−7.79 (m, 2H), 7.66 (d, J = 6.95 Hz, 1H), 7.44 (s, 1H), 7.34−7.31 (m, 2H), 7.30−7.28 (m, 2H), 7.27−7.26 (m, 3H), 7.25−7.24 (m, 4H), 7.23−7.20 (m, 6H), 7.18 (m, 1H), 6.51 (d, J = 1.55, Hz, 1H), 2.39 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.9, 145.6, 143.1, 136.7, 135.2, 133.8, 133.3, 129.3, 128.8, 128.7, 128.4, 128.2, 128.1, 128.0, 127.8, 126.3, 123.5, 115.6, 113.8, 100.5, 21.4 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{27}N_2]^+$: 475.2174; found: 475.2170.

8-Methoxy-2-phenyl-3-(1,3,3-triphenylpropa-1,2-dien-1-yl)imidazo[1,2-a]pyridine ***3r***. Following the general experimental procedure **I**, compound **3r** was obtained (using 10% EtOAc/petroleum ether) as brown solid (220 mg, 85%); mp: 175−176 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.93−7.91 (m, 2H), 7.44 (d, J = 6.80 Hz, 1H), 7.35−7.33 (m, 2H), 7.31−7.29 (m, 1H), 7.27−7.26 (m, 5H), 7.26−7.25 (m, 2H), 7.25−7.23 (m, 5H), 7.20−7.15 (m, 3H), 6.57−6.54 (m, 1H), 6.47 (d, J = 7.50 Hz, 1H), 4.05 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.8, 149.2, 143.1, 139.8, 135.3, 133.9, 133.7, 129.3, 129.0, 128.8, 128.7, 128.2, 128.1, 128.0, 127.8, 127.5, 126.3, 117.2, 115.2, 113.8, 112.3, 101.1, 100.9, 56.0 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{27}N_2O]^+$: 491.2123; found: 491.2121; IR (film): ν_{max} 2920, 1684, 1547, 1444, 1279, 1071, 1045, 751, 692, 665 cm^{-1}.

6-Bromo-2-phenyl-3-(1,3,3-triphenylpropa-1,2-dien-1-yl)-imidazo[1,2-a]pyridine ***3s***.[12] Following the general experimental procedure **I**, compound **3s** was obtained (using 20% EtOAc/petroleum ether) as white solid (259 mg, 91%); mp: 186−187 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.04−7.03 (m, 2H), 6.72 (d, J = 9.45 Hz, 1H), 6.47−6.46 (m, 5H), 6.46−6.45 (m, 4H), 6.44−6.43 (m, 5H), 6.40−6.38 (m, 6H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 209.7, 144.5, 143.8, 135.1, 133.4, 129.3, 128.8 (2), 128.4, 128.3 (2), 128.2, 128.0, 127.9, 126.3, 124.4, 118.2, 114.7, 114.5, 107.2, 100.5 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{24}BrN_2]^+$ and $[M + 2 + H]^+$ calculated for $[C_{34}H_{25}BrN_2]^{2+}$: 540.1190; found: 540.1193.

2-(4-(Trifluoromethyl)phenyl)-3-(1,3,3-triphenylpropa-1,2-dien-1-yl)imidazo[1,2-a] pyridine ***3t***.[12] Following the general experimental procedure **I**, compound **3t** was obtained (using 20% EtOAc/petroleum ether) as a solid (245 mg, 88%); mp: 165−166 °C ^1H NMR (500 MHz, $CDCl_3$): δ 7.96 (d, J = 8.30 Hz, 2H), 7.81 (d, J = 6.85 Hz, 1H), 7.71 (d, J = 9.05 Hz, 1H), 7.39 (d, J = 8.40 Hz, 2H), 7.34−7.33 (m, 3H), 7.29−7.27 (m, 4H), 7.25−7.24 (m, 4H), 7.21−7.20 (m, 5H), 6.70 (td, J = 6.80, 0.95 Hz, 1H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.7, 145.6, 142.4, 137.4, 134.9, 133.5, 129.5, 128.7 (2), 128.4, 128.3, 128.0, 126.2, 125.3 (2), 124.4, 117.8, 115.4, 114.0, 112.9, 100.4 ppm; ^{19}F (470 MHz, $CDCl_3$): −62.5; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{24}F_3N_2]^+$: 529.1892; found: 529.1897.

2-(4-Bromophenyl)-3-(1,3,3-triphenylpropa-1,2-dien-1-yl)imidazo[1,2-a]pyridine ***3u***. Following the general experimental procedure **I**, compound **3u** was obtained (using 20% EtOAc/petroleum ether) as white solid (250 mg, 88%); mp: 242−243 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.78 (d, J = 6.85 Hz, 1H), 7.72−7.68 (m, 3H), 7.33−7.32 (m, 4H), 7.30−7.28

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(m, 6H), 7.27 (m, 1H), 7.25 (m, 2H), 7.22−7.20 (m, 5H), 6.68 (td, J = 6.85, 1.10 Hz, 1H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.7, 145.5, 142.8, 135.0, 133.6, 132.7, 131.5, 129.5, 129.4, 128.7, 128.4, 128.2, 126.2, 125.2, 124.3, 121.9, 117.6, 114.6, 113.9, 112.7, 100.5 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{24}BrN_2]^+$: 539.1123; found: 539.1119.

2-(p-Tolyl)-3-(1,3,3-triphenylpropa-1,2-dien-1-yl)imidazo-[1,2-a]pyridine ***3v***.[12] Following the general experimental procedure **I**, compound **3v** was obtained (using 20% EtOAc/petroleum ether) as white solid (226 mg, 90%); mp: 286−287 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.78−7.76 (m, 2H), 7.68 (d, J = 9.05 Hz, 1H), 7.36−7.34 (m, 2H), 7.31−7.28 (m, 3H), 7.27−7.25 (m, 5H), 7.25−7.24 (m, 6H), 7.18−7.15 (m, 1H), 7.00 (d, J = 8.30 Hz, 2H), 6.63 (td, J = 6.75, 1.05 Hz, 1H), 2.31 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.9, 145.3, 143.9, 137.5, 135.3, 133.9, 130.8, 129.3, 129.2, 128.8, 128.6, 128.2, 128.0, 127.9, 126.3, 124.9, 124.2, 117.5, 113.9, 113.8, 112.4, 100.8, 21.3 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{27}N_2]^+$: 475.2174; found: 475.2170.

2-(4-Methoxyphenyl)-3-(1,3,3-triphenylpropa-1,2-dien-1-yl)imidazo[1,2-a]pyridine ***3w***.[12] Following the general experimental procedure **I**, compound **3w** was obtained (using 10% EtOAc/petroleum ether) as white solid (233 mg, 90%); mp: 149−150 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.82−7.80 (m, 2H), 7.76 (d, J = 6.85 Hz, 1H), 7.66 (d, J = 9.00 Hz, 1H), 7.36−7.34 (m, 2H), 7.32−7.30 (m, 2H), 7.28−7.27 (m, 6H), 7.25 (m, 5H), 7.18−7.15 (m, 1H), 6.74−6.72 (m, 2H), 6.65−6.62 (m, 1H), 3.77 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.9, 159.4, 145.3, 143.7, 135.3, 133.8, 129.3 (2), 128.8, 128.7, 128.2, 128.0, 126.4, 126.3, 124.8, 124.2, 117.3, 113.8, 113.7, 113.4, 112.4, 100.8, 55.3 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{27}N_2O]^+$: 491.2123; found: 491.2129; IR (film): ν_{max} 1609, 1547, 1501, 1488, 1030, 829, 786, 742, 728, 692 cm^{-1}.

2-(Naphthalen-2-yl)-3-(1,3,3-triphenylpropa-1,2-dien-1-yl)imidazo[1,2-a]pyridine ***3x***.[12] Following the general experimental procedure **I**, compound **3x** was obtained (using 10% EtOAc/petroleum ether) as white solid (232 mg, 86%); mp: 281−282 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.44 (s, 1H), 7.99 (d, J = 8.50 Hz, 1H), 7.82 (d, J = 6.80 Hz, 1H), 7.75 (t, J = 7.05 Hz, 2H), 7.67−7.63 (m, 2H), 7.44−7.39 (m, 3H), 7.33−7.30 (m, 3H), 7.22−7.21 (m, 7H), 7.16−7.14 (m, 5H), 6.68 (t, J = 6.70 Hz, 1H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.7, 145.5, 142.8, 135.0, 133.6, 132.7, 131.5, 129.5, 129.4, 128.7, 128.4, 128.2, 126.2, 125.2, 124.3, 121.9, 117.6, 114.6, 113.9, 112.7, 100.5 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{38}H_{27}N_2]^+$: 511.2174; found; 511.2176.

3-(3,3-Diphenyl-1-(p-tolyl)propa-1,2-dien-1-yl)-7-methyl-2-phenylimidazo[1,2-a]pyridine ***3y***. Following the general experimental procedure **I**, compound **3y** was obtained (using 10% EtOAc/petroleum ether) as brown solid (221 mg, 87%); mp: 185−186 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.88−7.86 (m, 2H), 7.65 (d, J = 6.95 Hz, 1H), 7.46 (s, 1H), 7.27−7.26 (m, 2H), 7.25 (m, 4H), 7.24−7.22 (m, 6H), 7.20−7.19 (m, 3H), 7.10 (d, J = 8.20 Hz, 2H), 6.48 (d, J = 6.95 Hz, 1H), 2.38 (s, 3H), 2.32 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 209.6, 145.7, 138.2, 136.0, 135.5, 133.9, 130.9, 130.0, 128.8, 128.6, 128.4, 128.0, 127.6, 126.2, 123.5, 115.9, 115.2, 113.8, 113.6, 100.6, 96.2, 21.5, 21.3 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{36}H_{29}N_2]^+$: 489.2331; found: 489.2335; IR (film): ν_{max} 2919, 2851, 1646, 1503, 1443, 1383, 1241, 1027, 906, 729, 693 cm^{-1}.

6-Bromo-3-(3,3-diphenyl-1-(4-(trifluoromethyl)phenyl)-propa-1,2-dien-1-yl)-2-phenylimidazo[1,2-a]pyridine ***3z***. Following the general experimental procedure **I**, compound **3z** was obtained (using 20% EtOAc/petroleum ether) as white solid (203 mg, 85%); mp: 177−178 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.88−7.84 (m, 3H), 7.63 (d, J = 9.45 Hz, 1H), 7.54 (d, J = 8.30 Hz, 2H), 7.43 (d, J = 8.20 Hz, 2H), 7.35−7.34 (m, 6H), 7.30−7.28 (m, 5H), 7.26−7.24 (m, 4H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 210.6, 144.4, 143.8, 137.4, 134.5, 132.9, 130.4, 130.1, 129.0, 128.8, 128.7, 128.6, 128.4, 127.9, 126.5, 125.1, 124.3, 123.0, 118.2, 115.3, 113.9, 107.7, 99.7 ppm; ^{19}F (470 MHz, $CDCl_3$): −62.5; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{35}H_{23}BrF_3N_2]^+$: 607.0997; found; 607.0996.

3-(3,3-Diphenyl-1-(4-(trifluoromethyl)phenyl)propa-1,2-dien-1-yl)-2-(p-tolyl)imidazo[1,2-a]pyridine ***3aa***. Following the general experimental procedure **I**, compound **3aa** was obtained (using 10% EtOAc/petroleum ether) as brown solid (196 mg, 85%); mp: 51−52 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.73−7.72 (m, 4H), 7.55 (d, J = 8.30 Hz, 2H), 7.44 (d, J = 8.30 Hz, 2H), 7.33−7.27 (m, 6H), 7.25−7.21 (m, 5H), 7.01 (d, J = 7.90 Hz, 2H), 6.68 (t, J = 6.80 Hz, 1H), 2.31 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 210.7, 145.4, 144.1, 137.8, 134.7, 130.5, 130.2, 129.9, 129.2, 128.8 (2), 128.4, 127.8, 126.5, 126.3 (2), 125.2, 124.0, 123.0, 117.5, 114.4, 113.1, 112.8, 100.0, 21.3 ppm; ^{19}F (470 MHz, $CDCl_3$): −62.5; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{36}H_{26}F_3N_2]^+$: 543.2048; found: 543.2045; IR (film): ν_{max} 1614, 1492, 1386, 1321, 1166, 1122, 1067, 907, 767, 695 cm^{-1}.

3-(1-([1,1′-Biphenyl]-4-yl)-3,3-diphenylpropa-1,2-dien-1-yl)-2-(p-tolyl)imidazo[1,2-a]pyridine ***3ab***. Following the general experimental procedure **I**, compound **3ab** was obtained (using 20% EtOAc/petroleum ether) as brown gummy product (194 mg, 82%); ^1H NMR (100 MHz, $CDCl_3$): δ 8.42 (d, J = 9.00 Hz, 1H), 8.05 (d, J = 6.80 Hz, 1H), 7.82 (t, J = 7.80 Hz, 1H), 7.66 (d, J = 8.05 Hz, 2H), 7.61 (d, J = 8.30 Hz, 2H), 7.56 (d, J = 7.45 Hz, 2H), 7.46−7,38 (m, 5H), 7.37−7.33 (m, 3H), 7.31−7.28 (m, 4H), 7.19 (m, 4H), 7.11 (d, J = 7.90 Hz, 2H), 2.35 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (500 MHz, $CDCl_3$): δ 210.5, 141.8, 141.4, 140.4, 140.0, 137.7, 134.4, 131.2, 131.0, 130.0, 129.0, 128.9, 128.8, 128.7, 128.4, 127.9, 127.0, 126.4, 125.0, 124.7, 116.4, 115.6, 115.5, 115.4, 98.0, 21.5 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{41}H_{31}N_2]^+$: 551.2487; found: 551.2486.

6-Bromo-2-phenyl-3-(1-phenyl-3,3-di-p-tolylpropa-1,2-dien-1-yl)imidazo[1,2-a]pyridine ***3ac***. Following the general experimental procedure **I**, compound **3ac** was obtained (using 10% EtOAc/petroleum ether) as white solid (246 mg, 90%); mp: 143−144 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 7.92−7.90 (m, 2H), 7.87 (s, 1H), 7.57 (d, J = 9.45 Hz, 1H), 7.32−7.28 (m, 4H), 7.25−7.21 (m, 5H), 7.17 (d, J = 8.00 Hz, 4H), 7.11 (d, J = 7.90 Hz, 4H), 2.36 (s, 6H) ppm; $^{13}C\{H\}$ NMR (125 MHz, $CDCl_3$): δ 209.7, 144.3, 143.8, 138.2, 133.8, 133.5, 132.2, 129.5, 129.3, 128.7, 128.5, 128.3, 128.2, 127.9, 126.3, 124.6, 118.1, 115.1, 114.4, 107.2, 100.2, 21.3 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{36}H_{28}BrN_2]^+$: 567.1436; found: 567.1437.

6-Bromo-3-(1,3-diphenylbuta-1,2-dien-1-yl)-2-phenylimidazo[1,2-a]pyridine ***3ad***. Following the general experimental procedure **I**, compound **3ad** was obtained (using 10% EtOAc/petroleum ether) as brown gummy product (195

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mg, 83%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.95 (s, 1H), 7.82 (d, J = 6.80 Hz, 2H), 7.69 (d, J = 7.05 Hz, 1H), 7.58−7.53 (m, 1H), 7.24 (m, 2H), 7.23 (m, 2H), 7.21 (m, 2H), 7.20 (m, 2H), 7.18 (m, 2H), 6.77−6.71 (m, 2H), 5.19−5.11 (m, 1H), 2.06 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 208.8, 144.5, 143.6, 135.4, 134.0, 133.6, 129.3, 129.2, 128.8(2), 128.5, 128.3, 128.2, 128.1, 127.9, 127.6, 127.1, 126.5, 126.3(2), 125.9, 124.2, 121.1, 118.3, 117.6, 115.6, 107.3, 105.6, 99.1, 16.42 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{29}H_{22}BrN_2]^+$: 477.0966; found; 477.0968.

7-Methyl-2-phenyl-3-(3-phenyl-1-(p-tolyl)prop-2-yn-1-yl)-imidazo[1,2-a]pyridine ***6***. Following the general experimental procedure **II**, compound **6** was obtained (using 30% EtOAc/petroleum ether) as brown liquid product (236 mg, 85%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.89 (d, J = 7.00 Hz, 1H), 7.77−7.75 (m, 2H), 7.47 (t, J = 7.35 Hz, 2H), 7.44−7.42 (m, 3H), 7.41−7.38 (m, 1H), 7.31−7.30 (m, 4H), 7.29 (s, 1H), 7.12 (d, J = 8 Hz, 2H), 7.52 (d, J = 1.35, Hz, 1H), 6.07 (s, 1H), 2.37 (s, 3H), 2.32 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 145.8, 143.8, 137.3, 135.8, 134.3, 133.9, 131.9, 129.7, 128.8(2), 128.5, 128.4, 128.1, 127.0, 124.4, 122.9, 117.0, 116.0, 114.7, 85.9, 85.1, 32.9, 21.4, 21.1 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{30}H_{24}N_2]^+$: 413.2018; found: 413.2017.

3-(1,5-Diphenyl-3-(p-tolyl)pent-2-en-4-yn-1-yl)-2-phenylimidazo[1,2-a]pyridine ***8a***. Following the general experimental procedure **II**, compound **8a** was obtained (using 30% EtOAc/petroleum ether) as brown gummy product (206 mg, 89%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.78−7.75 (m, 3H), 7.70 (d, J = 9.05 Hz, 1H), 7.50 (d, J = 8.20 Hz, 2H), 7.33−7.31 (m, 3H), 7.30−7.29 (m, 2H), 7.28−7.27 (m, 2H), 7.25−7.23 (m, 3H), 7.21 (s, 1H), 7.20−7.18 (m, 1H), 7.15 (m, 1H), 7.16−7.15 (m, 2H), 7.14 (s, 1H), 6.87 (d, J = 9.75 Hz, 1H), 6.67 (td, J = 6.85, 1.20 Hz, 1H), 6.52 (d, J = 5.00 Hz, 1H), 2.35 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 145.1, 145.0, 139.1, 138.3, 134.7, 134.6, 131.8, 131.7, 129.2, 129.2, 128.6(2), 128.2, 127.8, 127.3, 127.2, 126.9, 126.4, 126.3, 126.0, 124.5, 124.3, 122.7, 119.7, 118.0, 112.3, 97.0, 85.9, 42.3, 21.2 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{37}H_{29}N_2]^+$: 501.2331; found: 501.2334.

3-(1,5-Diphenyl-3-(p-tolyl)pent-2-en-4-yn-1-yl)-7-methyl-2-phenylimidazo[1,2-a] pyridine ***8b***. Following the general experimental procedure **II**, compound **8b** was obtained (using 30% EtOAc/petroleum ether) as brown gummy product (214 mg, 90%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.78−7.76 (m, 2H), 7.63 (d, J = 7.00 Hz, 1H), 7.49 (d, J = 8.20 Hz, 3H), 7.32 (s, 1H), 7.30 (s, 2H), 7.29−7.28 (m, 2H), 7.27 (m, 1H), 7.24 (d, J = 1.7 Hz, 2H), 7.22 (m, 2H), 7.21 (s, 1H), 7.17 (d, J = 1.25 Hz, 1H), 7.15 (d, J = 1.35 Hz, 2H), 7.14 (s, 1H), 6.84 (d, J = 9.75 Hz, 1H), 6.51 (dd, J = 1.45, 1.40 Hz, 1H), 6.34 (d, J = 9.70 Hz, 1H), 2.38 (s, 3H), 2.35 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 145.3, 139.2, 138.3, 134.7, 131.8 (2), 129.2 (2), 129.1, 128.6 (2), 128.2, 127.8, 127.3, 127.2, 126.9, 126.4, 123.8, 122.7, 119.1, 116.2, 115.1, 96.9, 85.8, 42.3, 21.4, 21.2 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{38}H_{31}N_2]^+$: 515.2487; found: 515.2488.

6-Bromo-3-(1,5-diphenyl-3-(p-tolyl)pent-1-en-4-yn-1-yl)-2-phenylimidazo[1,2-a]pyridine ***8c***. Following the general experimental procedure **II**, compound **8c** was obtained (using 30% EtOAc/petroleum ether) as brown gummy product (236 mg, 88%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.93 (s, 1H), 7.72 (d, J = 6.76 Hz, 2H), 7.60 (d, J = 9.48 Hz, 1H), 7.47 (d, J = 8.16 Hz, 2H), 7.35−7.33 (m, 2H), 7.31 (m, 2H), 7.29−7.28 (m, 2H), 7.26 (s, 1H), 7.24 (m, 2H), 7.22−7.21 (m, 2H), 7.19 (s, 1H), 7.17−7.14 (m, 4H), 6.80 (d, J = 9.52 Hz, 1H), 6.31 (d, J = 9.48 Hz, 1H), 2.35 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 145.4, 143.3, 138.6, 138.5, 134.5, 133.7, 131.7, 131.4, 129.3(2), 129.1, 128.7, 128.3, 128.2, 128.1, 127.5, 127.2(2), 126.4, 124.3, 122.5, 120.6, 118.2, 107.3, 97.3, 85.7, 42.3, 21.2 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{37}H_{28}BrN_2]^+$: 579.1430; found: 579.1432.

3-(1,5-Diphenyl-3-(p-tolyl)pent-1-en-4-yn-1-yl)-2-(4-trifluoromethyl)phenyl)imidazo[1,2-a]pyridine ***8d***. Following the general experimental procedure **II**, compound **8d** was obtained (using 30% EtOAc/petroleum ether) as yellowish gel (234 mg, 89%); ^{1}H NMR (500 MHz, $CDCl_3$): δ ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.84 (d, J = 8.1 Hz, 2H), 7.79 (d, J = 6.9 Hz, 1H), 7.69 (d, J = 8.1 Hz, 2H), 7.54 (d, J = 8.25 Hz, 2H), 7.48 (d, J = 8.1 Hz, 2H), 7.34−7.30 (m, 3H), 7.28 (d, J = 6.8 Hz, 1H), 7.25−7.24 (m, 2H), 7.23 (m, 1H), 7.21−7.19 (m, 2H), 7.15−7.14 (m, 4H), 6.83 (d, J = 9.65 Hz, 1H), 6.71 (t, J = 9.65 Hz, 1H), 6.31 (d, J = 9.65 Hz, 1H), 2.35 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 145.3, 143.6, 138.8, 138.5, 138.3, 134.5, 131.5, 131.4, 129.5, 129.4, 129.3, 128.8, 128.3, 127.4, 127.2, 127.2, 126.3, 125.02 (q, J_{C-F} = 15.5 Hz), 124.7, 122.4, 120.5, 118.2, 112.5, 97.2, 85.6, 42.3, 21.2 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{38}H_{28}F_3N_2]^+$: 569.2199; found: 569.2195.

3-(1,5-Diphenyl-3-(p-tolyl)pent-1-en-4-yn-1-yl)-2-(4-methoxyphenyl)imidazo[1,2-a]pyridine ***8e***. Following the general experimental procedure **II**, compound **8e** was obtained (using 30% EtOAc/petroleum ether) as brown gummy product (193 mg, 86%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.74−7.68 (m, 4H), 7.52 (d, J = 8.20 Hz, 2H), 7.38−7.34 (m, 1H), 7.32−7.29 (m, 2H), 7.28−7.25 (m, 3H), 7.24−7.19 (m, 3H), 7.18−7.14 (m, 4H), 6.89 (d, J = 9.90 Hz, 1H), 6.83 (d, J = 8.80 Hz, 2H), 6.64 (td, J = 6.80, 1.15 Hz, 1H), 6.36 (d, J = 9.85 Hz, 1H), 3.67 (s, 3H), 2.35 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 159.4, 145.1, 145.0, 139.1, 138.3, 134.8, 131.8, 131.7, 130.3, 129.2 (2), 128.5, 128.2, 127.3, 127.2, 126.8, 126.4, 124.5, 124.1, 122.7, 119.1, 117.9, 114.0, 112.0, 96.8, 86.0, 55.2, 42.3, 21.3 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{38}H_{31}N_2O]^+$: 531.2431; found: 531.2430.

3-(3-(4-Ethoxyphenyl)-1,5-diphenylpent-2-en-4-yn-1-yl)-2-(4-methoxyphenyl)imidazo[1,2-a]pyridine ***8f***. Following the general experimental procedure **II**, compound **8f** was obtained (using 30% EtOAc/petroleum ether) as brown gummy product (206 mg, 87%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.69−7.67 (m, 2H), 7.64−7.62 (d, J = 8.70 Hz, 2H), 7.48 (d, J = 8.80 Hz, 1H), 7.30−7.28 (m, 2H), 7.24 (s, 1H), 7.22 (d, J = 4.25 Hz, 1H), 7.20 (s, 1H), 7.19−7.17 (m, 4H), 7.15 (s, 1H), 7.13 (s, 1H), 7.12−7.10 (m, 2H), 6.79 (d, J = 8.800 Hz, 2H), 6.75 (d, J = 9.00 Hz, 2H), 6.72 (s, 1H), 6.61 (t, J = 6.75 Hz, 1H), 6.27 (d, J = 9.80 Hz, 1H), 3.96 (q, J = 6.95 Hz, 2H), 3.60 (s, 3H), 1.33 (t, J = 7.00 Hz, 1H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 159.6, 159.3, 139.0, 131.8, 130.4, 130.2, 129.8, 129.2, 128.6, 128.2, 127.7, 127.3, 127.2, 126.6, 124.6, 122.7, 119.5, 117.4, 114.5, 114.1, 112.6, 96.9, 85.9, 63.6, 55.2, 42.3, 14.9 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{39}H_{32}N_2O_2]^+$: 561.2542; found: 561.2542.

3-(3-(2-Bromophenyl)-1,5-diphenylpent-2-en-4-yn-1-yl)-2-(4-methoxyphenyl)imidazo[1,2-a]pyridine ***8g***. Following the general experimental procedure **II**, compound **8g** was obtained (using 30% EtOAc/petroleum ether) as brown gummy product (195 mg, 85%); ^{1}H NMR (400 MHz,

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$CDCl_3$): δ 7.70−7.66 (m, 2H), 7.59 (d, J = 6.92 Hz, 1H), 7.53 (d, J = 7.60 Hz, 1H), 7.46−7.45 (m, 1H), 7.32−7.30 (m, 1H), 7.26 (t, J = 1.28 Hz, 1H), 7.23 (d, J = 6.08 Hz, 2H), 7.22−7.20 (m, 2H), 7.18 (s, 3H), 7.17 (m, 1H), 7.14 (m, 1H), 7.12−7.09 (m, 1H), 7.08−7.06 (m, 2H), 6.78 (d, J = 8.84 Hz, 2H), 6.55 (dd, J = 1.12, 1.12 Hz, 1H), 6.51 (d, J = 9.84 Hz, 1H), 6.26 (d, J = 9.6 Hz, 1H), 3.59 (s, 3H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 159.7, 145.2, 144.7, 139.9, 138.3, 137.3, 133.3, 131.8, 130.8, 130.5, 130.3, 129.5, 129.2, 128.9, 128.6, 128.2(2), 127.6, 127.3, 124.7, 122.6, 122.5, 118.8, 117.5, 114.2, 112.6, 97.47, 85.5, 55.2, 41.9 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{37}H_{28}BrN_2O]^+$: 595.1385; found: 595.1389.

4-(3-(3-Methoxyphenyl)-1,5-diphenylpent-2-en-4-yn-1-yl)-n,n-dimethylaniline ***9***. Following the general experimental procedure **II**, compound **9** was obtained (using 35% EtOAc/petroleum ether) as brown liquid product (170 mg, 87%); ^1H NMR (500 MHz, $CDCl_3$): δ 7.53−7.51 (m, 2H), 7.35−7.33 (m, 3H), 7.31−7.30 (m, 4H), 7.29−7.27 (m, 1H), 7.26−7.24 (m, 2H), 7.23−7.20 (m, 1H), 7.15 (d, J = 8.50 Hz, 2H), 6.87−6.85 (d, J = 10.00 Hz, 1H), 6.84−6.82 (m, 1H), 6.71 (d, J = 8.8 Hz, 2H), 5.52 (d, J = 10.15 Hz, 1H), 3.82 (s, 3H), 2.91 (s, 6H) ppm; $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 159.7, 149.4, 144.0, 140.2, 139.5, 131.7, 131.3, 129.4, 129.1, 128.5(2), 128.4(2), 126.4, 123.4, 123.1, 118.8, 113.1, 112.9, 112.3, 95.4, 86.9, 55.3, 51.4, 40.7 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{32}H_{30}NO]^+$: 444.2322; found: 444.2322.

*2-Iodo-1,3,3-triphenyl-1-(2-phenylimidazo[1,2-a]pyridin-3-yl)prop-2-en-1-ol (****10****)*. Following the general experimental procedure **III**, compound **10** was obtained from **3a** (using 50% EtOAc/petroleum ether) as brown liquid product (134 mg, 68%); ^1H NMR (500 MHz, $CDCl_3$): δ 8.67 (d, J = 6.9 Hz, 1H), 7.55 (d, J = 8.8 Hz, 1H), 7.35 (t, J = 7.35 Hz, 3H), 7.25−7.23 (m, 5H), 7.11−7.10 (m, 1H), 7.06 (m, 2H), 7.01−6.98 (m, 6H), 6.96−6.95 (m, 3H), 6.84 (bs, 2H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 152.2, 149.6, 144.1, 141.8, 139.4, 129.4, 129.0, 128.8, 128.7, 128.4, 128.1, 127.8, 127.7, 127.6, 127.4, 127.0, 126.6, 126.5, 125.2, 117.4, 112.2, 81.7 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{26}IN_2O]^+$: 605.1090; found; 605.1095.

1-(6-Bromo-2-phenylimidazo[1,2-a]pyridin-3-yl)-2-iodo-1,3,3-triphenylprop-2-en-1-ol ***11***. Following the general experimental procedure **III**, compound **11** was obtained from **3s** (using 50% EtOAc/petroleum ether) as brown liquid product (134 mg, 70%); ^1H NMR (500 MHz, $CDCl_3$): δ 8.75 (s, 1H), 7.30 (t, J = 7.65 Hz, 4H), 7.26−7.25 (m, 2H), 7.22−7.18 (m, 4H), 7.05−7.03 (m, 2H), 6.97 (m, 2H), 6.95−6.94 (m, 4H), 6.91−6.87 (m, 4H), 6.83 (s, 1H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 152.8, 149.5, 145.0, 142.4, 141.1, 139.3, 134.7, 128.9, 128.6, 128.5, 128.2, 128.0, 127.8, 127.5, 127.1, 126.8, 126.6, 117.7, 106.8, 81.6 ppm; HRMS (ESI-TOF) m/z: $[M + H]^+$ calculated for $[C_{34}H_{25}BrIN_2O]^+$: 683.0195; found: 683.0148.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.5c00326.

Experimental procedures and characterization data for all reactions and products, including copies of copies of ^1H and $^{13}C\{H\}$ NMR spectra and details of single-crystal data of compounds **3a** (CCDC: 2408426) (PDF)

Accession Codes

Deposition Number 2408426 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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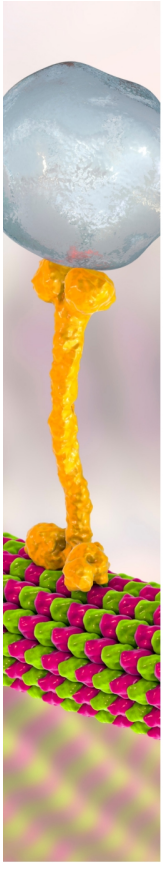
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